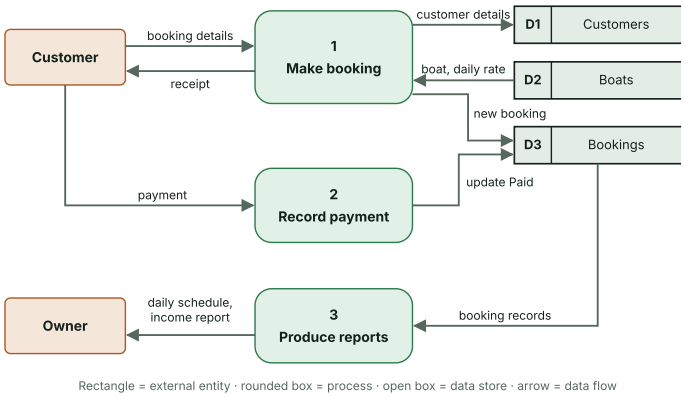
Analysis: client, current system and requirements

AD02 FACT FILE

Specification: students should be able to

- Use appropriate fact-finding methods, including interviewing the client, to investigate a problem
- Describe the current system, its data, processes and problems
- Produce a requirements specification with numbered, measurable functional and non-functional requirements
- Model the proposed system using data flow diagrams

Key words: fact-finding, interview, current system, requirements specification, functional requirement, non-functional requirement, measurable, acceptance criteria, data flow diagram, external entity, data store, context diagram



A level 1 data flow diagram for the proposed Erne Boat Hire system with two external entities, three processes and three data stores

Fact-finding

Analysis begins with fact-finding. Interviews give depth and allow follow-up questions but are time-consuming and reflect one person's view. Questionnaires collect views from many users quickly but cannot be clarified. Observation shows how work really happens, though people may behave differently when watched. Examining existing documents such as booking sheets reveals the actual data items, formats and volumes. In the A2 2 task the case study supplies most facts, and a recorded interview with the client, often role-played by the teacher, adds detail.

Requirements specification

A requirements specification is a numbered list of what the new system must do and how well it must do it. Functional requirements describe data to be stored, processes such as calculations and searches, and outputs such as reports. Non-functional requirements describe qualities such as usability, performance, security and reliability. Each requirement should be specific and measurable, with clear acceptance criteria, so it can be tested and later evaluated as fully, partially or not met. Vague words such as 'easy', 'fast' and 'efficient' need numbers attached.

Data flow diagrams

A data flow diagram (DFD) models how data moves through a system. External entities are sources or destinations of data outside the system. Processes transform data and are numbered. Data stores hold data at rest, and data flows are labelled arrows. A context diagram shows the whole system as a single process with its external entities. A level 1 DFD expands this into the main processes and data stores. Data cannot flow directly between two data stores or between two external entities without a process.

Key terms from the mark schemes

Requirements specification
A numbered, agreed list of the functional and non-functional requirements that the new system must meet.

Functional requirement
A statement of what the system must do, such as the data it stores, the processes it carries out or the outputs it produces.

Non-functional requirement
A statement of how well the system must perform, covering qualities such as usability, performance, security or reliability.

Measurable requirement
A requirement with a specific, testable target so that it can be judged objectively as met or not met.

Data flow diagram
A diagram that models how data moves between external entities, processes and data stores in a system.

Context diagram
A top-level data flow diagram showing the whole system as one process together with its external entities and data flows.

Exam focus

Analysis evidence should be specific to the client in the case study, not general theory. Show the interview questions and answers, then show how each finding became a requirement. Number every requirement and make it testable: if you cannot imagine a test with an expected result, rewrite it. Include non-functional requirements with measurable targets. Make sure DFD labels are meaningful nouns for data flows and verb phrases for processes, and that every flow connects to a process. The same requirement numbers must reappear in testing and evaluation.

? Check yourself

1. Distinguish between a functional and a non-functional requirement, giving an example of each for a boat hire booking system. [4]
2. Explain why the requirement 'The system should be user-friendly' is weak and rewrite it. [3]
3. Give one advantage and one disadvantage of interviewing the client as a fact-finding method. [2]

Answers: 1. A functional requirement states what the system must do – the data, processes and outputs (1) / e.g. the system must calculate the total cost as number of days × daily rate of the boat (1) / A non-functional requirement states how well the system must perform – usability, performance, security, reliability (1) / e.g. only staff with a valid username and password can open the booking forms, or any form reachable from the main menu in no more than two clicks (1). 2. It is vague / subjective – 'user-friendly' means different things to different people (1) / so it cannot be measured or tested objectively, and cannot be evaluated as met or not met (1) / rewritten, e.g.: 'A new member of staff can add a booking in under 2 minutes after a 10-minute demonstration' or 'Every form can be reached from the main menu in no more than two clicks' (1). 3. Advantage, any one: allows follow-up questions to clarify answers / gives detailed, in-depth information about the client's needs (1) / Disadvantage, any one: time-consuming to arrange, carry out and write up / gives only one person's view, which may be biased or incomplete (1).